Chapter 1: Learning Algorithms Overview

Part II — Introduction to Statistical Learning

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Abstract

Part I gave us the language of probability – sample spaces, random variables, moments. Part II turns that language into *learning algorithms*: procedures that look at data and produce a function, a partition, or a policy. This opening chapter is a map of the territory: the handful of learning paradigms (supervised, unsupervised, semi-supervised, online, reinforcement, graph-based, and distillation-based) that every later chapter in this part is an instance of. Read it as a table of contents with explanations, not as a self-contained technical chapter – each paradigm gets its full mathematical treatment later (regression and cross-validation in Chapter 2, unsupervised learning and PCA in Chapter 5, and so on).

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1 Supervised Learning

Definition 1.1. In supervised learning we observe *labeled* training data $\{(x_i, y_i)\}_{i=1}^{n}$, where x_i is an instance and y_i its label, and we assume a generating model

$$y = f(x) + \epsilon,$$

where ϵ is unobserved noise. A learning algorithm maps the training set to an estimate $\hat{f}(\cdot)$ of f. The goal is that $\hat{f}(x)$ is close to y for *new* (x, y) pairs, not just the ones we trained on – this distinction (fit vs. generalize) is the entire subject of Chapter 2 (bias–variance

tradeoff, cross-validation).

Example 1.1 (Advertising)**.** X = advertising budget (\$000s), Y = sales (000s of units); each data point is one past advertising campaign. Fitting a line to this scatterplot by least squares is the simplest possible supervised learning algorithm – formalized fully in Chapter 2.

Example 1.2 (Image classification – CIFAR-10)**.** X = a 32×32 color image, $Y \in \{0, \ldots, 9\}$ one of ten object classes. The same supervised-learning definition applies; only f (now a convolutional network, Part II Chapter 9) and the loss function (cross- entropy, not squared error) change. (Krizhevsky & Hinton, 2009.)

2 Unsupervised Learning

Definition 2.1. In unsupervised learning we observe *unlabeled* data $x_1, \ldots, x_n$ (no y_i) and ask what structure – clusters, a lower-dimensional representation, a density – the data itself reveals.

Example 2.1 (Clustering)**.** Given unlabeled points, K-means with $K = 3$ partitions them into three groups by proximity to three learned centroids (full treatment: Part II, Chapter 7).

Example 2.2 (Learned image compression)**.** Image compression represents an image with as few bits as possible. Learned, probability-model-based codecs have been shown to beat classical codecs (BPG, JPEG, JPEG2000) on perceptual quality at the same bit rate (Mentzer et al., 2018) – an unsupervised (or self-supervised) learning problem: there is no external label, only the image itself to reconstruct.

Example 2.3 (Masked language modeling)**.** BERT (Devlin et al., 2018) is trained by masking tokens in a sentence and predicting them from context – the "label" is manufactured from the unlabeled text itself. This blurs the line between unsupervised and supervised learning, which is exactly why the next paradigm exists.

3 Semi-Supervised Learning

Definition 3.1. Semi-supervised learning uses a small labeled dataset together with a much larger unlabeled dataset, typically by having a model trained on the labeled data generate "pseudo-labels" for the unlabeled data, which are then used for further training.

Example 3.1. Billion-scale semi-supervised image classification: a teacher model labels a huge unlabeled image corpus; a student model trains on the combined labeled + pseudo-labeled data (Yalniz et al., 2019).

3.1 Further Examples Worth Knowing

- **Deep belief networks** for handwritten-digit classification and generation – an early (pre-2012) example of layerwise unsupervised pretraining feeding into supervised fine-tuning.
- **Reinforcement learning** milestones: emergent tool-use in multi-agent hide-and-seek,

and AlphaGo / AlphaZero learning to play Go (and chess/shogi) purely from self-play.

4 Online Learning

Ordinary supervised/unsupervised learning assumes: data arrives as one static batch; the model is trained once; the model is then frozen. Three problems follow directly from that assumption:

- it is not well-suited to genuinely streaming data;
- retraining from scratch on the full history is wasteful once the history is large;
- the fitted model becomes stale as the data-generating process drifts over time (*concept drift*).

Definition 4.1. *Online (incremental) learning* algorithms update their parameters one new observation (or mini-batch) at a time, without revisiting the full history, so the model can track a data stream as it evolves.

The simplest possible instance is an incremental (running) mean: given the current mean $\bar{x}_{n-1}$ over $n-1$ points and a new observation x_n,

$$\bar{x}_n = \bar{x}_{n-1} + \frac{x_n - \bar{x}_{n-1}}{n},$$

which is mathematically identical to the batch mean but needs $O(1)$ memory and work per new point instead of $O(n)$. Exercise 1 asks you to implement exactly this update (Putatunda, 2021).

5 Reinforcement Learning

Definition 5.1. In reinforcement learning an *agent* interacts with an *environment* over a sequence of time steps, choosing actions to maximize cumulative reward, with no explicit (x_i, y_i) pairs telling it the "correct" action – only a reward signal, often delayed. This is covered here only as a paradigm; a full treatment (Markov decision processes, policy gradients, multi-armed bandits) is planned for a future part of this book.

6 Graph Representation Learning

Definition 6.1. When data is naturally a graph (a social network, a molecule, a knowledge graph), graph representation learning produces vector embeddings of nodes (or whole graphs) that preserve graph structure, so that ordinary supervised or unsupervised algorithms can be applied downstream. This paradigm is also deferred to a full treatment in Part III.

7 Knowledge Distillation

Definition 7.1. In *knowledge distillation* (teacher/student learning), a large "teacher" model's output distribution (not just its hard labels) is used as a training target for a smaller "student" model, transferring the teacher's learned function into a cheaper model to deploy. This is a supervised-learning technique whose "labels" are themselves the output of another learned model – a nice illustration of how the paradigms in this chapter combine rather than partition the field.

8 Summary

Summary

- **Supervised**: learn $\hat{f}$ from labeled pairs (x_i, y_i).
- **Unsupervised**: find structure in unlabeled x_i alone.
- **Semi-supervised**: combine a little labeled data with a lot of unlabeled data via pseudo-labeling.
- **Online**: update the model incrementally as data streams in, to track concept drift.
- **Reinforcement**: learn a policy from reward signals, not labels.
- **Graph representation learning**: embed graph-structured data for downstream use.
- **Knowledge distillation**: train a small model to imitate a large one's output distribution.

These are not mutually exclusive categories – masked language modeling (BERT) and knowledge distillation both blend supervised mechanics onto unsupervised or model-generated targets. The rest of Part II works through each paradigm's core algorithms in full mathematical detail.

9 Exercises

See `exercises/python/exercise01.py`, `exercises/r/exercise01.R`, and `exercises/rust/exercise01.rs`: a closed-form ordinary-least- -squares fit (supervised), a single K-means assignment step (unsupervised), and the incremental-mean update (online learning) – one small, working example of each of the three paradigms most fully developed later in Part II. The broken counterparts to debug are in `exercise01_buggy.{py,R,rs}`.

References

- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2018). BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv:1810.04805*.
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- Yalniz, I. Z., Jégou, H., Chen, K., Paluri, M., & Mahajan, D. (2019). Billion-scale semi-supervised learning for image classification. *arXiv:1905.00546*.